

Experiment No. 2 — Logistic Regression

Logistic Regression is a **Supervised Machine Learning Classification Algorithm** used to predict **categories/classes**.

Unlike Linear Regression:

- Linear Regression → predicts continuous values
- Logistic Regression → predicts classes/categories

1. Real-Life Examples

Logistic Regression is used for:

Input	Output
Study Hours	Pass / Fail
Email Content	Spam / Not Spam
Tumor Features	Cancer / No Cancer
Customer Data	Buy / Not Buy

2. What is Logistic Regression?

It predicts the probability of a class.

Example:

- 0 → Fail
- 1 → Pass

The output probability ranges between:

0 to 1

Sigmoid Function

Logistic Regression uses the sigmoid function:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

This converts output into probability between:

- 0 and 1

3. Difference Between Linear and Logistic Regression

Linear Regression	Logistic Regression
Predicts continuous values	Predicts categories
Output can be any number	Output between 0 and 1
Used for regression	Used for classification

4. Workflow

```
Dataset
↓
Preprocessing
↓
Train-Test Split
↓
Train Logistic Model
↓
Prediction
↓
Evaluation
```

5. Required Libraries

```
pip install pandas numpy matplotlib scikit-learn
```

6. Import Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix
```

7. Create Sample Dataset

We will predict:

- Pass = 1
- Fail = 0

based on study hours.

```
data = {
    'Hours_Studied': [1, 2, 3, 4, 5, 6, 7, 8],
    'Pass': [0, 0, 0, 0, 1, 1, 1, 1]
}

df = pd.DataFrame(data)

print(df)
```

Dataset Explanation

Hours Studied Pass

1	0
2	0
7	1
8	1

8. Define Features and Target

```
X = df[['Hours_Studied']]
y = df['Pass']
```

9. Split Dataset

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

10. Train Logistic Regression Model

```
model = LogisticRegression()  
model.fit(X_train, y_train)
```

What `.fit()` Does

The model learns:

- relationship between study hours and passing probability.
-

11. Predict Results

```
y_pred = model.predict(X_test)  
print("Predicted Values:")  
print(y_pred)
```

Example Output

```
[1 0]
```

Means:

- Student 1 → Pass
 - Student 2 → Fail
-

12. Evaluate the Model

Accuracy Score

Measures correct predictions.

Formula:

$$\text{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Predictions}}$$

```
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)
```

13. Confusion Matrix

Used to evaluate classification models.

```
cm = confusion_matrix(y_test, y_pred)
print("Confusion Matrix:")
print(cm)
```

Confusion Matrix Structure

Actual / Predicted	0	1
0	TN	FP
1	FN	TP

Where:

- TP → True Positive
 - TN → True Negative
 - FP → False Positive
 - FN → False Negative
-

14. Predict New Data

Example:

- Student studied 5 hours

```
new_data = [[5]]
prediction = model.predict(new_data)
print("Prediction:", prediction[0])
```

Output Meaning

1 → Pass
0 → Fail

15. Visualization

```
plt.scatter(df['Hours_Studied'], df['Pass'], color='blue')

plt.xlabel("Hours Studied")
plt.ylabel("Pass / Fail")
plt.title("Logistic Regression")

plt.show()
```

16. Complete Final Code

```
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix

# Dataset
data = {
    'Hours_Studied': [1, 2, 3, 4, 5, 6, 7, 8],
    'Pass': [0, 0, 0, 0, 1, 1, 1, 1]
}

df = pd.DataFrame(data)

# Features and Target
X = df[['Hours_Studied']]
y = df['Pass']

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Train model
model = LogisticRegression()

model.fit(X_train, y_train)
```

```
# Prediction
y_pred = model.predict(X_test)

print("Predicted Values:")
print(y_pred)

# Accuracy
accuracy = accuracy_score(y_test, y_pred)

print("Accuracy:", accuracy)

# Confusion Matrix
cm = confusion_matrix(y_test, y_pred)

print("Confusion Matrix:")
print(cm)

# Predict new value
new_data = [[5]]

prediction = model.predict(new_data)

print("Prediction for 5 study hours:", prediction[0])

# Visualization
plt.scatter(df['Hours_Studied'], df['Pass'], color='blue')

plt.xlabel("Hours Studied")
plt.ylabel("Pass / Fail")
plt.title("Logistic Regression")

plt.show()
```

17. Viva Questions

Q1. What is Logistic Regression?

Logistic Regression is a supervised classification algorithm used to predict categorical outcomes.

Q2. What is the output range of Logistic Regression?

0 to 1

Q3. Which function is used in Logistic Regression?

Sigmoid Function.

Q4. Difference between Linear and Logistic Regression?

- Linear → continuous output
- Logistic → categorical output

Q5. What is Accuracy?

Accuracy measures the percentage of correct predictions.

18. Short Conclusion

Thus, the Logistic Regression model was successfully implemented to classify data into different classes. The model was trained, tested, and evaluated using accuracy score and confusion matrix.